11.2.1 Need for Graphics Standards

During the early period of computer graphics, only a few organizations could afford the computer capabilities required for interactive computer graphics. Now-a-days, improving technology and reducing costs have made graphics affordable to more and more users. The growing popularity of the graphics created a demand for standards to avoid duplication. It was also necessary to make application programs portable to new machines, and to build on work of others. As the new graphics softwares developed, common features began to appear, and certain concepts became accepted practice. These software packages and the commonality among their features led to graphics standards.

On the other hand, we have a number of output devices such as plotters, printers, cathode ray tubes, liquid crystal displays projection systems etc. All these devices have provided many technical ways of creating images with different performance and cost ranges. To face such diversity, device independence is one goal of standardization. Therefore much of the computer graphics standards deal with a set of common concepts that can be used across an interestingly wide range of applications and hardware. The various differences among the devices are simple variations. They can be accommodated within a suitably defined set of standards.

The standards govern the methods for connecting different graphics hardware to the computer. It also deals with the internal construction of the software. They impinge as little as possible on the behaviour of the application program as seen by the user. On the other hand it is the programmers who write interactive applications who must become familiar with the workings of the standards in order to exploit them.

11.2.2 Graphics Standard

Several graphics standards have been developed or are under development. The various standards differ in the kinds of services they provide and their role in a graphics system. Some of the types of graphics standards are as follows :

Interface Standards : These standardize the communication at an interface between two parts of a hardware/software system. Infact all the standards are included under this type.

Data Interface Standards : They specify the digital representation of graphical data. The digital data is used to send commands from a computer to a graphics display or to store pictures in a data file.

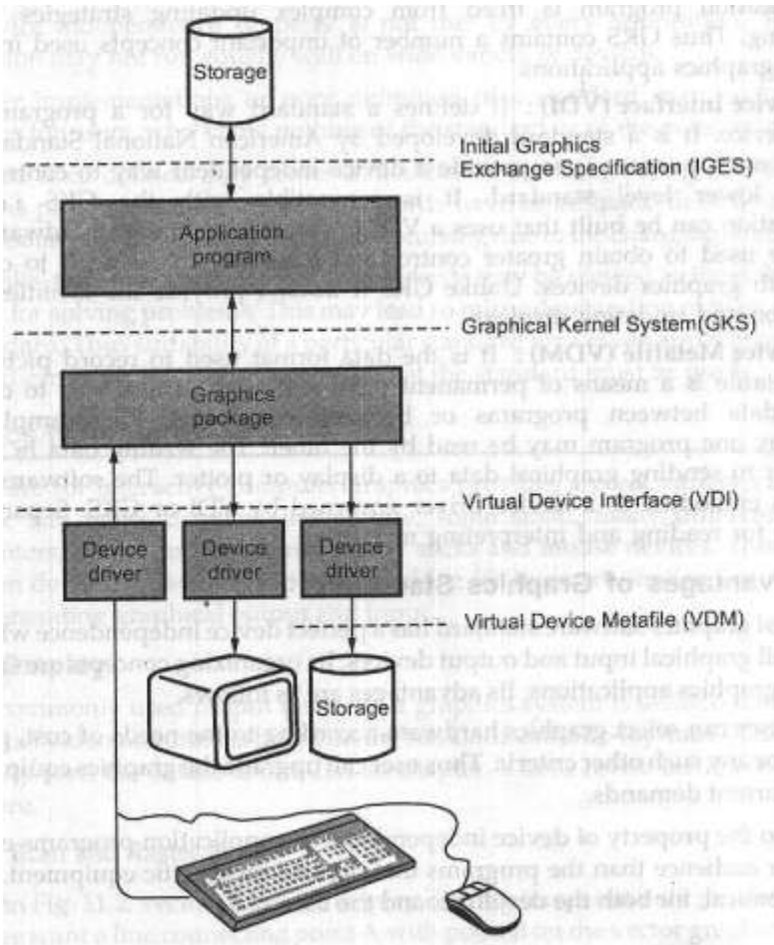

Fig. 11.1 Graphics standard used in application program

Subroutine Interface Standards : These standards specify the behaviour of a set of subroutines that can be called to manipulate graphical data.

Fig. 11.1 shows how graphics standards may be used by an application program. Various parts of the system are explained below.

The Initial Graphics Exchange Standard (IGES) : The IGES defines a format for engineering drawing that can be read or written by applications. It allows graphical engineering databases to be transferred between applications.

Graphical Kernel System (GKS) : The GKS is a subroutine interface that application programmers use to manage graphical input/output. It is an international standard, approved by the International Standards Organisation (ISO). It is the most fully developed of the graphics standards. It specifies the interface to a graphics subroutine library, which addresses the graphical input/output needs of a wide range of applications. It facilitates device independent graphical output and input, for structuring, maintaining, storing and retrieving pictures. The picture manipulation facilities of GKS can be used to keep a display image up-to-date as changes are made. Thus application program is freed from complex updating strategies and tedious book-keeping. Thus GKS contains a number of important concepts used in writing the interactive graphics applications.

Virtual Device Interface (VDI) : It defines a standard way for a program to drive a graphics device. It is a standard developed by American National Standards Institute (ANSI). Its main purpose is to provide a device independent way to control hardware. VDI is a lower level standard. It is compatible with the GKS i.e. the GKS implementation can be built that uses a VDI to communicate with hardware devices. It can also be used to obtain greater control and flexibility by using it to communicate directly with graphics devices. Unlike GKS it doesn't provide the facilities for picture manipulation and updating displays.

Virtual Device Metafile (VDM) : It is the data format used to record pictures on disk files. A metafile is a means of permanent picture storage. It is a way to communicate graphical data between programs or between computers, for example a picture produced by one program may be read by the other. The writing data in a metafile is very similar to sending graphical data to a display or plotter. The software for creating metafiles is embodied in a device driver governed by VDI or GKS. Separate software is available for reading and interpreting metafiles.

1. The **3D Core Graphics System** (a.k.a. **Core**) was the very first graphical standard ever developed. A group of 25 experts of the ACM Special Interest Group SIGGRAPH developed this "conceptual framework". The specifications were published in 1977 and it became a foundation for many future developments in the field of computer graphics.
2. **PHIGS** (**Programmer's Hierarchical Interactive Graphics System**) is an API standard for rendering 3D computer graphics, considered to be the 3D graphics standard for the 1980s through the early 1990s.
3. The **Initial Graphics Exchange Specification (IGES)** (pronounced *eye-jess*) is a file format which defines a vendor neutral data format that allows the digital exchange of information among Computer-aided design (CAD) systems.
4. The **Graphical Kernel System** (**GKS**) was the first ISO standard for low-level computer graphics, introduced in 1977. GKS provides a set of drawing features for two-dimensional vector graphics suitable for charting and similar duties. The calls are designed to be portable across different

programming languages, graphics devices and hardware, so that applications written to use GKS will be readily portable to many platforms and devices.

11.2.3 Advantages of Graphics Standards

The ideal graphics software standard has a perfect device independence which allows it to operate all graphical input and output devices. Its organizing concepts are suitable for all interactive graphics applications. Its advantages are as follows.

1. The user can select graphics hardware according to the needs of cost, performance, size, or any such other criteria. Thus user can upgrade the graphics equipment to fulfil the current demands.
2. Due to the property of device independence the application programs can address a wider audience than the programs that depend on specific equipment. This is very economical, for both the developer and the user.
3. The standard software packages that provide necessary facilities can be used. Thus everytime development of new graphics routines can be avoided.
4. A standard programming language and graphics standards facilitate the transporting of an application program from one computer to another, having compiler of the given programming language and implementing same graphics standards.
5. It also facilitates the training of application programmers who use interactive graphics. By learning standard, a programmer will be able to write interactive programs for any computer that has an implementation of the standards. This property is known as **programmer portability**.

11.2.4 Hazards of Graphics Standards

Despite of their greate values, graphics standards present certain hazards. The main reason behind this is the fact that an ideal standard does not exist.

1. The device independence property is not met by every application. The given application may not run equally well on wide variety of devices.
2. Improper implementations or poor definition of a standard may lead to create problems for users, who know nothing of the standard or of the potential problem.
3. Standards once defined may create difficulties in accommodating new needs. To avoid this problem to some extent standards have to be made flexible enough to accommodate new needs and conditions occurring due to the changing environment.
4. By mistake, the methods embodied in standards may be viewed as the only suitable method for solving problems. This may lead to misunderstanding or even misuse of the standard. Thus suitability of a particular standard for a given application must be analysed carefully. It is not an assertion that the standard must be used.